

Max Duration: 90 mins

PART B

Max Marks: 35

- Answer all the questions of this part precisely and neatly in the answer booklet.
- For all numerical questions, provide all the steps clearly with the units mentioned.

1. [10] Answer the following questions briefly in not more than 2-3 sentences.

- Mention three distinct features of a software-defined network.
- How is the data plane implementation different in SDN compared to traditional switching? How can SDN be used to implement destination-based forwarding?
- Mention any two characteristics of Google WAN that motivated the use of SDN?
- What motivated the use of centralized TE solution for Google WAN?
- Mention any two issues that prevent extending SDN to any general WAN.

2. [7] (a) Consider a 100Mbps Ethernet, on which a malicious node detects transmission and disrupts it by beginning a competing transmission as soon as it hears the beginning of the transmitted frame, thereby causing a collision. Suppose your node detects collision during the transmission of its 12th bit on the wire. Given that the speed of the signal in the wire is 2×10^8 m/s, compute the distance of the malicious node (in m)?

(b) What is the advantage of using exponential backoff in CSMA/CD compared to random backoff?

3. [8] Consider four autonomous systems A, B, C, and D connected to each other in a linear topology A - B - C - D. Suppose A receives a BGP path announcement to D with AS_PATH as "{B,C,D}". Answer the following questions briefly.

- If A is given to be the provider of B, deduce the possible relationships for "B-C" and "C-D".
- If A is given to be customer of B, deduce the possible relationships for "B-C" and "C-D".
- If A and B are known to be peers, how would your answer change?

4. [10] Consider a file of size say, n bytes (multiple of 1000 bytes) being transferred from a source to a destination connected through a chain topology consisting of 5 switches. Each link has a bandwidth of 4 Mbps and a propagation delay of 2 ms. Assume that the switches can support both packet switching and circuit switching. So, there are two options for file transfer. In the first option, you send the file using 1KB packets which includes a header of 24 bytes. Each packet also incurs a processing time of 1 ms at every switch. In the second option, you set up a circuit by sending a 1 KB control message in both directions after which you transfer the file continuously. The processing delay for the setup message at each switch is also 1 ms but, there is no delay during the data transfer.

- For what file size (of n bytes) is the total number of bytes sent across the network less for circuit switching than that for packet switching?
- For what file size (of n bytes) is the total latency incurred before the entire file is transmitted, smaller for circuit switching than that for packet switching?